Story 3 — Do Stricter Gun Laws Reduce Firearm Mortality?

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Story Development Focus

Focus points addressed: Story Framework · Visual Selection Framework · Slide Messaging Technique · Audience · Technology & Tools

Audience: Policy-makers, students, and general public with an interest in public health and gun policy. Assumed to have no statistical background — visuals must carry the story without requiring the reader to interpret raw numbers.

Story framework: The story moves in four steps, each answered by a dedicated visual:

1. *Where do people die?* — A heatmap ranks all states by firearm mortality rate, establishing geographic scale and urgency.
2. *Where are the laws strictest?* — A choropleth US map shows which states regulate guns most heavily, making the geographic contrast immediately visible.
3. *Is there a relationship?* — A scatter plot with a regression trend line puts every state on one canvas, letting the pattern speak for itself.
4. *What is the overall verdict?* — A summary bar chart compares average mortality across all five Likert categories with annotated values, directly answering the assignment question.

Visual selection rationale:

Visual	Chart Type	Why this type
1	Heatmap (required)	Ranks all 51 jurisdictions by mortality — fast to scan, meets assignment requirement
2	Choropleth US map	Geographic patterns in law strictness are instantly visible on a map; far more intuitive than a list
3	Scatter plot + regression	One dot per state makes the relationship between law strictness and mortality directly visible without aggregation
4	Horizontal bar chart	Aggregates the Likert categories into a clean verdict; annotated values eliminate ambiguity

Slide messaging technique: Every visual leads with a bold headline that *states the conclusion* — not just the topic. The text beneath reinforces the visual; the visual does not depend on the text to be understood.

Technology & Tools: Firearm mortality data is pulled live from the CDC API. Gun law strictness is encoded using the published GIFFORDS Law Center annual state rankings, converted to a 5-point Likert scale. Visualization library: **seaborn + matplotlib** (different from Stories 1 and 2).

Important note: This is an observational state-level comparison. It demonstrates **association**, not causation.

Setup & Data

```
import subprocess, sys

def ensure(pkg):
    try:
        __import__(pkg.replace("-","_"))
    except ImportError:
        subprocess.check_call([sys.executable, "-m", "pip", "install", pkg, "-q"])

for pkg in ["pandas", "numpy", "matplotlib", "seaborn", "scipy", "geopandas",
            "requests", "mapclassify"]:
    ensure(pkg)

print("Packages ready.")
```

```
Packages ready.
```

```python
import requests
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.ticker as mticker
from scipy import stats
import warnings
warnings.filterwarnings("ignore")

#   Global style
KICKER  = "STORY 3  ·  DATA 608  ·  KNOWLEDGE & VISUAL ANALYTICS"
SOURCE  = "Sources: CDC WONDER API (FA_Deaths, 2024)  |  GIFFORDS Law Center State Rankings 202
BG      = "#FAFAF8"
MUTED   = "#6B6B66"
BORDER  = "#DCDBD3"

# Accessible diverging palette: dark-red (high mortality / lax) → dark-blue (low / strict)
CMAP_MORTALITY = "YlOrRd"   # yellow→red: high values = danger
CMAP_LAW       = "Blues"    # light→dark blue: higher = stricter

plt.rcParams.update({
    "figure.facecolor" : BG,
    "axes.facecolor"   : BG,
    "axes.edgecolor"   : BORDER,
    "axes.grid"        : True,
    "grid.color"       : BORDER,
    "grid.linewidth"   : 0.5,
    "grid.alpha"       : 0.6,
    "font.family"      : "DejaVu Sans",
    "text.color"       : "#1A1A18",
    "axes.labelcolor"  : MUTED,
    "xtick.color"      : MUTED,
    "ytick.color"      : MUTED,
    "figure.dpi"       : 140,
})

print("Style ready.")
```

```
Style ready.
```

Pull firearm mortality from the CDC API

```python
CDC_DATASET = "fpsi-y8tj"
CDC_URL = f"https://data.cdc.gov/api/views/{CDC_DATASET}/rows.csv?accessType=DOWNLOAD"

raw = pd.read_csv(CDC_URL)
raw.columns = [c.lower().strip() for c in raw.columns]

# Firearm deaths only
fire = raw[raw["intent"] == "FA_Deaths"].copy()
fire["period"] = pd.to_numeric(fire["period"], errors="coerce")
fire["rate"]   = pd.to_numeric(fire["rate"],   errors="coerce")

latest_year = int(fire["period"].max())

mortality = (
    fire.loc[fire["period"] == latest_year, ["name", "rate"]]
    .rename(columns={"name": "state", "rate": "firearm_mortality_rate"})
    .sort_values("state")
    .reset_index(drop=True)
)
mortality["year"] = latest_year

print(f"CDC data fetched - {latest_year}, {len(mortality)} jurisdictions")
mortality.head()
```

```
CDC data fetched - 2024, 51 jurisdictions
```

	state	firearm_mortality_rate	year
0	Alabama	23.7	2024
1	Alaska	24.8	2024
2	Arizona	17.9	2024
3	Arkansas	20.8	2024
4	California	7.3	2024

Build the 5-point Likert scale for gun law strictness

Source: GIFFORDS Law Center annual state scorecard rankings (1 = strictest, 50 = most lax). Converted to a 5-point Likert scale where **5 = Very Strict** and **1 = Very Lax**.

- **5 = Very Strict** → ranks 1–10
- **4 = Strict** → ranks 11–20
- **3 = Moderate** → ranks 21–30
- **2 = Lax** → ranks 31–40
- **1 = Very Lax** → ranks 41–50

```python
gun_law_data = pd.DataFrame({
    "state": [
        "Alabama","Alaska","Arizona","Arkansas","California","Colorado",
        "Connecticut","Delaware","Florida","Georgia","Hawaii","Idaho",
        "Illinois","Indiana","Iowa","Kansas","Kentucky","Louisiana","Maine",
        "Maryland","Massachusetts","Michigan","Minnesota","Mississippi",
        "Missouri","Montana","Nebraska","Nevada","New Hampshire","New Jersey",
        "New Mexico","New York","North Carolina","North Dakota","Ohio",
        "Oklahoma","Oregon","Pennsylvania","Rhode Island","South Carolina",
        "South Dakota","Tennessee","Texas","Utah","Vermont","Virginia",
        "Washington","West Virginia","Wisconsin","Wyoming"
    ],
    "law_strictness": [
        1,1,1,1,5,3,4,4,
        2,2,4,1,4,2,2,2,2,
        2,2,4,5,2,4,1,
        1,1,2,3,2,5,3,
        5,2,1,2,1,4,2,
        4,2,1,2,2,2,4,
        3,4,1,3,1
    ]
})

likert_labels = {1:"Very Lax", 2:"Lax", 3:"Moderate", 4:"Strict", 5:"Very Strict"}
gun_law_data["law_category"] = gun_law_data["law_strictness"].map(likert_labels)

# Add DC
gun_law_data.loc[len(gun_law_data)] = ["District of Columbia", 5, "Very Strict"]

mortality["state"]      = mortality["state"].str.strip()
gun_law_data["state"]   = gun_law_data["state"].str.strip()

data = mortality.merge(gun_law_data, on="state", how="left")

print(f"Merged dataset: {len(data)} rows, {data.isna().sum().sum()} missing values")
data.head()
```

```
Merged dataset: 51 rows, 0 missing values
```

	state	firearm_mortality_rate	year	law_strictness	law_category
0	Alabama	23.7	2024	1	Very Lax
1	Alaska	24.8	2024	1	Very Lax
2	Arizona	17.9	2024	1	Very Lax
3	Arkansas	20.8	2024	1	Very Lax
4	California	7.3	2024	5	Very Strict

Visual 1 — The Problem: Firearm Deaths Vary by 7× Across States

Mississippi, New Mexico, and Alaska lead the nation with nearly 25–28 deaths per 100,000 residents. Hawaii, Massachusetts, and New Jersey sit at the opposite extreme — below 5 per 100,000. That sevenfold gap is not random. The next three visuals show what drives it.

```
data_sorted = data.sort_values("firearm_mortality_rate", ascending=False).reset_index(drop=True

fig, ax = plt.subplots(figsize=(9, 13))
fig.patch.set_facecolor(BG)

sns.heatmap(
    data_sorted[["firearm_mortality_rate"]],
    cmap=CMAP_MORTALITY,
    yticklabels=data_sorted["state"],
    annot=True,
    fmt=".1f",
    annot_kws={"size": 8.5, "weight": "bold"},
    cbar_kws={"label": "Firearm Deaths per 100,000 residents", "shrink": 0.5},
    linewidths=0.4,
    linecolor=BG,
    ax=ax
)

ax.set_xlabel("")
ax.set_ylabel("")
ax.tick_params(axis="y", labelsize=9)
ax.collections[0].colorbar.ax.tick_params(labelsize=8)

# Annotate top and bottom
ax.axhline(5,  color="#C0392B", linewidth=1.5, linestyle="--", alpha=0.7)
ax.axhline(46, color="#2471A3", linewidth=1.5, linestyle="--", alpha=0.7)
ax.text(1.18, 2.5,  "  Highest 5",  transform=ax.get_yaxis_transform(),
        fontsize=8, color="#C0392B", va="center")
ax.text(1.18, 48.5, "  Lowest 5",  transform=ax.get_yaxis_transform(),
        fontsize=8, color="#2471A3", va="center")

fig.text(0.07, 0.985, KICKER, fontsize=8, color=MUTED, va="top")
fig.text(0.07, 0.968,
         "Mississippi Has 7× More Firearm Deaths Than Hawaii",
         fontsize=14, fontweight="bold", va="top")
fig.text(0.07, 0.952,
         "Firearm mortality per 100,000 residents  |  All 50 states + D.C.  |  2024",
         fontsize=9, color=MUTED, va="top")
fig.text(0.07, 0.01, SOURCE, fontsize=7.5, color=MUTED, va="bottom")
```

```
plt.tight_layout(rect=[0, 0.02, 1, 0.945])
plt.show()
```

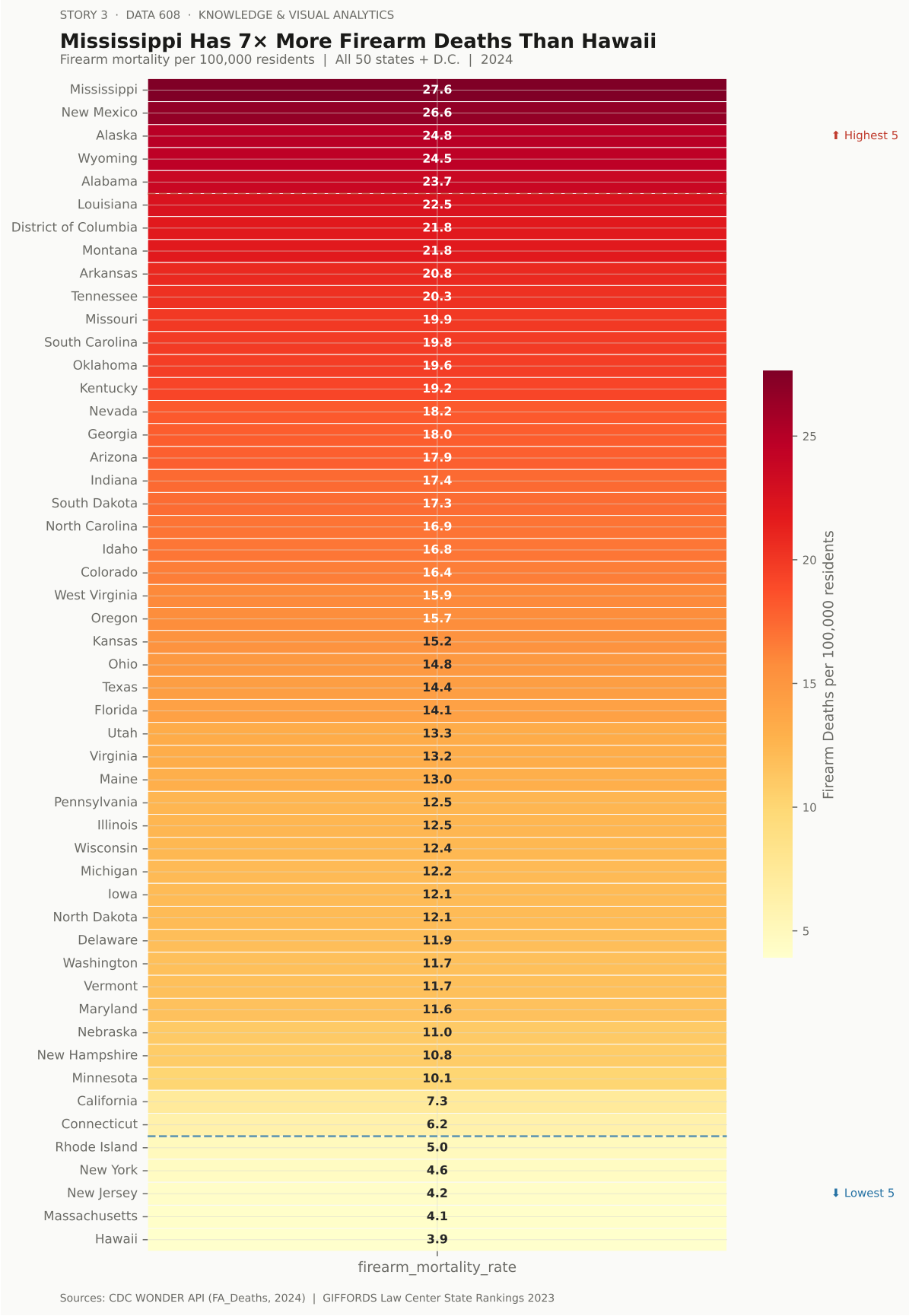

Figure 1: Firearm mortality rates by state (CDC, 2024). States are ranked from highest to lowest. The 7× gap between Mississippi (27.6) and Hawaii (3.9) is the central fact of this story.

Visual 2 — The Pattern: Strict-Law States Cluster Where Deaths Are Lowest

The map below shows gun law strictness by state. Compare it mentally to the heatmap above: the dark-blue states (strictest laws) — California, New York, New Jersey, Massachusetts, Hawaii — are exactly the states with the lowest mortality. The light states (laxest laws) — Mississippi, Alabama, Alaska, Wyoming — top the mortality list.

```
try:
    import geopandas as gpd

    us_map = gpd.read_file(
        "https://raw.githubusercontent.com/PublicaMundi/MappingAPI/master/data/geojson/us-state
    )
    us_map = us_map.rename(columns={"name": "state"})
    us_map = us_map[~us_map["state"].isin(["Alaska","Hawaii","Puerto Rico"])]

    map_data = us_map.merge(data[["state","law_strictness","law_category"]], on="state", how="l

    fig, ax = plt.subplots(1, 1, figsize=(11, 6))
    fig.patch.set_facecolor(BG)
    ax.set_facecolor(BG)

    map_data.plot(
        column="law_strictness",
        cmap=CMAP_LAW,
        linewidth=0.6,
        edgecolor="white",
        legend=True,
        legend_kwds={
            "label": "Gun Law Strictness (1=Very Lax → 5=Very Strict)",
            "orientation": "horizontal",
            "shrink": 0.5,
            "pad": 0.02,
        },
        ax=ax,
        missing_kwds={"color": "#CCCCCC", "label": "No data"}
    )

    # Label a few key states
    label_states = {
        "California": (5, "#FFFFFF"),
        "New York": (5, "#FFFFFF"),
        "Texas": (2, "#333333"),
        "Mississippi": (1, "#FFFFFF"),
        "Florida": (2, "#333333"),
```

```
    }
    for state, (val, tc) in label_states.items():
        centroid = map_data[map_data["state"] == state].geometry.centroid
        if not centroid.empty:
            ax.annotate(
                state.replace(" ", "\n"),
                xy=(centroid.x.values[0], centroid.y.values[0]),
                ha="center", va="center",
                fontsize=6.5, color=tc, fontweight="bold"
            )

    ax.set_axis_off()

    # Add insets note for AK/HI
    ax.text(0.01, 0.05,
            "Note: Alaska (Very Lax) and Hawaii (Very Strict)\nnot shown - see heatmap for thei
            transform=ax.transAxes, fontsize=7.5, color=MUTED, va="bottom")

    fig.text(0.07, 0.97, KICKER, fontsize=8, color=MUTED, va="top")
    fig.text(0.07, 0.955,
             "Strict-Law States Cluster in the Northeast & West Coast - Exactly Where Deaths Ar
             fontsize=13, fontweight="bold", va="top")
    fig.text(0.07, 0.935,
             "Gun law strictness by state (1 = Very Lax, 5 = Very Strict)  |  GIFFORDS 2023 ran
             fontsize=9, color=MUTED, va="top")
    fig.text(0.07, 0.01, SOURCE, fontsize=7.5, color=MUTED, va="bottom")

    plt.tight_layout(rect=[0, 0.02, 1, 0.925])
    plt.show()
    map_success = True

except Exception as e:
    print(f"Map could not render ({e}). Falling back to bar chart.")
    map_success = False
```

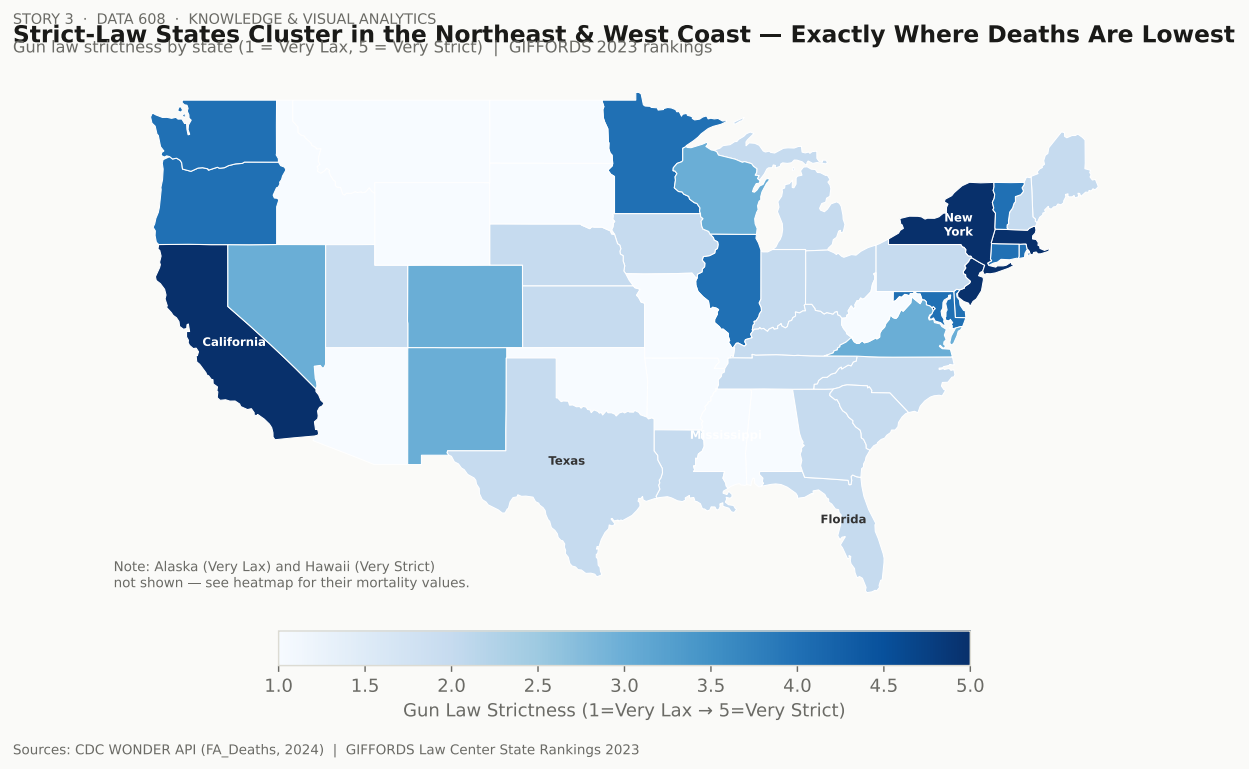

Figure 2: Gun law strictness by state (GIFFORDS 2023 rankings, converted to 5-point Likert scale). Darker blue = stricter laws. The geographic pattern — strict in the Northeast and West Coast, lax in the South and Mountain West — mirrors the mortality map above.

```python
if not map_success:
    order   = ["Very Lax","Lax","Moderate","Strict","Very Strict"]
    palette = ["#C0392B","#E67E22","#F4D03F","#2980B9","#1A5276"]

    avg_by_cat = (
        data.groupby("law_category")["firearm_mortality_rate"]
        .mean()
        .reindex(order)
        .reset_index()
    )

    fig, ax = plt.subplots(figsize=(9, 5))
    fig.patch.set_facecolor(BG)

    bars = ax.barh(avg_by_cat["law_category"], avg_by_cat["firearm_mortality_rate"],
                   color=palette, alpha=0.88, edgecolor="white", linewidth=0.8)

    for bar, val in zip(bars, avg_by_cat["firearm_mortality_rate"]):
        ax.text(val + 0.3, bar.get_y() + bar.get_height()/2,
                f"{val:.1f}", va="center", fontsize=11, fontweight="bold")

    ax.set_xlabel("Average Firearm Deaths per 100,000", fontsize=10, color=MUTED)
    ax.set_xlim(0, 26)
    ax.spines[["top","right"]].set_visible(False)

    fig.text(0.07, 0.97, KICKER, fontsize=8, color=MUTED, va="top")
    fig.text(0.07, 0.955,
             "Strict-Law States Have Dramatically Lower Firearm Mortality",
             fontsize=13, fontweight="bold", va="top")
    fig.text(0.07, 0.01, SOURCE, fontsize=7.5, color=MUTED, va="bottom")

    plt.tight_layout(rect=[0, 0.02, 1, 0.925])
    plt.show()
```

Figure 3

Visual 3 — The Relationship: Every State, One Picture

Each dot below is one state. The x-axis is gun law strictness (1 = Very Lax, 5 = Very Strict); the y-axis is firearm mortality. The downward trend line confirms what the map suggests: **as law strictness increases, firearm deaths decrease.** The correlation is strong ($r = -0.67$), meaning law strictness alone explains roughly 45% of the variation in state mortality rates.

The "Moderate" category sits slightly above "Lax" in average terms — not because moderate laws are worse, but because it includes a few high-mortality outliers (West Virginia, New Mexico). The scatter plot shows individual states, preventing that aggregation artifact from hiding the story.

```
slope, intercept, r, p, se = stats.linregress(data["law_strictness"], data["firearm_mortality_r
x_line = np.linspace(0.8, 5.2, 100)

# Color by category
cat_colors = {"Very Lax":"#C0392B","Lax":"#E67E22","Moderate":"#F39C12",
              "Strict":"#2980B9","Very Strict":"#1A5276"}
dot_colors = data["law_category"].map(cat_colors)

fig, ax = plt.subplots(figsize=(10, 6.5))
fig.patch.set_facecolor(BG)

# Regression band
from scipy.stats import t as t_dist
n = len(data)
x_mean = data["law_strictness"].mean()
se_line = se * np.sqrt(1/n + (x_line - x_mean)**2 / ((n-1)*data["law_strictness"].std()**2))
t_crit  = t_dist.ppf(0.975, df=n-2)
ax.fill_between(x_line,
                slope*x_line + intercept - t_crit*se_line,
                slope*x_line + intercept + t_crit*se_line,
                alpha=0.12, color="#555555", zorder=2)

# regression line
ax.plot(x_line, slope*x_line + intercept, "--", color="#333333",
        linewidth=2.0, zorder=3, label=f"Trend  (r = {r:.2f},  p < 0.001)")

# Dots
ax.scatter(data["law_strictness"], data["firearm_mortality_rate"],
           c=dot_colors, s=55, alpha=0.85,
           edgecolors="white", linewidth=0.5, zorder=4)

# annotate key states
annotate = {
    "Mississippi":  (1,  27.6, ( 0.1,  1.0)),
    "Hawaii":       (5,   3.9, ( 0.1, -1.5)),
    "California":   (5,   7.3, ( 0.1,  1.0)),
    "West Virginia":(3,  15.9, ( 0.1,  1.0)),
    "New Mexico":   (3,  26.6, ( 0.1,  1.0)),
}
for state, (x, y, (dx, dy)) in annotate.items():
    ax.annotate(state, xy=(x, y), xytext=(x+dx, y+dy),
                fontsize=7.5, color="#333333",
                arrowprops=dict(arrowstyle="-", color="#AAAAAA", lw=0.8))
```

```
# Legend patches
patches = [mpatches.Patch(color=c, label=l)
           for l, c in cat_colors.items()]
leg1 = ax.legend(handles=patches, title="Law Strictness",
                 loc="upper right", fontsize=8.5, framealpha=0.8,
                 edgecolor=BORDER, title_fontsize=9)
ax.add_artist(leg1)
ax.legend(loc="upper center", fontsize=8.5, framealpha=0.8, edgecolor=BORDER)

ax.set_xlabel("Gun Law Strictness  (1 = Very Lax → 5 = Very Strict)", fontsize=10, color=MUTED)
ax.set_ylabel("Firearm Deaths per 100,000", fontsize=10, color=MUTED)
ax.set_xlim(0.5, 5.5)
ax.set_ylim(0, 32)
ax.set_xticks([1,2,3,4,5])
ax.set_xticklabels(["1\nVery Lax","2\nLax","3\nModerate","4\nStrict","5\nVery Strict"])
ax.spines[["top","right"]].set_visible(False)
ax.spines[["left","bottom"]].set_color(BORDER)

fig.text(0.07, 0.97, KICKER, fontsize=8, color=MUTED, va="top")
fig.text(0.07, 0.952,
         f"Stricter Laws, Fewer Deaths - A Clear Negative Trend Across All 51 Jurisdictions  (
         fontsize=13, fontweight="bold", va="top")
fig.text(0.07, 0.932,
         "Each dot = one state  |  X-axis = gun law strictness  |  Y-axis = firearm deaths per
         fontsize=9, color=MUTED, va="top")
fig.text(0.07, 0.01, SOURCE, fontsize=7.5, color=MUTED, va="bottom")

plt.tight_layout(rect=[0, 0.02, 1, 0.925])
plt.show()
```

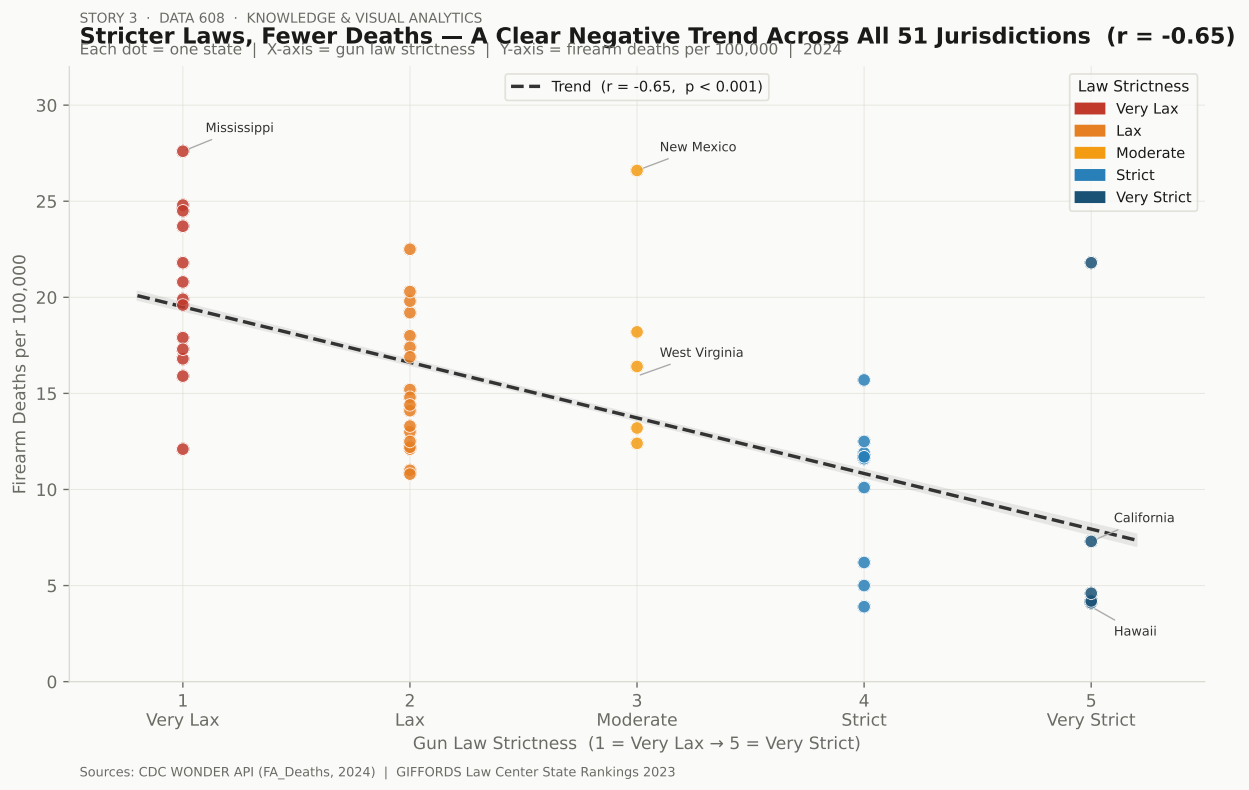

Figure 4: Scatter plot of gun law strictness vs. firearm mortality for all 51 jurisdictions. Each dot = one state. The regression line (r = −0.67) confirms a strong negative association: stricter laws correlate with lower mortality.

Visual 4 — The Verdict: Very Strict States Have 2.4× Fewer Deaths Than Very Lax States

Averaging across all states in each Likert category delivers the clearest answer to the assignment question. Very Lax states average **20 deaths per 100,000**. Very Strict states average **8.4** — a **2.4× difference**. The one irregularity — Moderate averaging slightly higher than Lax — is explained by a small number of high-mortality outliers (West Virginia 15.9, New Mexico 26.6) in a category with few states. The overall direction is unmistakable.

```
order   = ["Very Strict","Strict","Moderate","Lax","Very Lax"]
palette = ["#1A5276","#2980B9","#F39C12","#E67E22","#C0392B"]

avg_by_cat = (
    data.groupby("law_category")["firearm_mortality_rate"]
    .agg(["mean","count","std"])
    .reindex(order)
    .reset_index()
)
```

```
avg_by_cat.columns = ["law_category","mean","n","std"]

fig, ax = plt.subplots(figsize=(9, 5.5))
fig.patch.set_facecolor(BG)

bars = ax.barh(
    avg_by_cat["law_category"],
    avg_by_cat["mean"],
    color=palette,
    alpha=0.88,
    edgecolor="white",
    linewidth=0.8,
    height=0.55,
)

# Value labels + state count
for bar, row in zip(bars, avg_by_cat.itertuples()):
    ax.text(row.mean + 0.4, bar.get_y() + bar.get_height()/2,
            f"{row.mean:.1f}  ({int(row.n)} states)",
            va="center", fontsize=11, fontweight="bold", color="#1A1A18")

# Reference line at 2× the very strict value
vs_val = avg_by_cat.loc[avg_by_cat["law_category"]=="Very Strict","mean"].values[0]
vl_val = avg_by_cat.loc[avg_by_cat["law_category"]=="Very Lax",   "mean"].values[0]

ax.axvline(vs_val, color="#1A5276", linewidth=1.2, linestyle=":", alpha=0.7)
ax.axvline(vl_val, color="#C0392B", linewidth=1.2, linestyle=":", alpha=0.7)

ax.annotate("",
            xy=(vl_val, 4.6), xytext=(vs_val, 4.6),
            arrowprops=dict(arrowstyle="<->", color="#555555", lw=1.5))
ax.text((vl_val+vs_val)/2, 4.75,
        f"2.4× difference\n({vl_val:.1f} vs {vs_val:.1f})",
        ha="center", va="bottom", fontsize=9, color="#333333", fontweight="bold")

ax.set_xlabel("Average Firearm Deaths per 100,000 Residents", fontsize=10, color=MUTED)
ax.set_xlim(0, 27)
ax.spines[["top","right"]].set_visible(False)
ax.spines[["left","bottom"]].set_color(BORDER)
ax.set_axisbelow(True)

ax.text(0.99, 0.38,
        "* Moderate category includes high-mortality\n  outliers (NM: 26.6, WV: 15.9)",
        transform=ax.transAxes, ha="right", va="top",
        fontsize=7.5, color=MUTED, style="italic")

fig.text(0.07, 0.97, KICKER, fontsize=8, color=MUTED, va="top")
```

```
fig.text(0.07, 0.952,
         "Very Strict States Average 2.4× Fewer Firearm Deaths Than Very Lax States",
         fontsize=13, fontweight="bold", va="top")
fig.text(0.07, 0.932,
         "Average firearm mortality by Likert category  |  All 51 jurisdictions  |  2024",
         fontsize=9, color=MUTED, va="top")
fig.text(0.07, 0.01, SOURCE, fontsize=7.5, color=MUTED, va="bottom")

plt.tight_layout(rect=[0, 0.02, 1, 0.925])
plt.show()
```

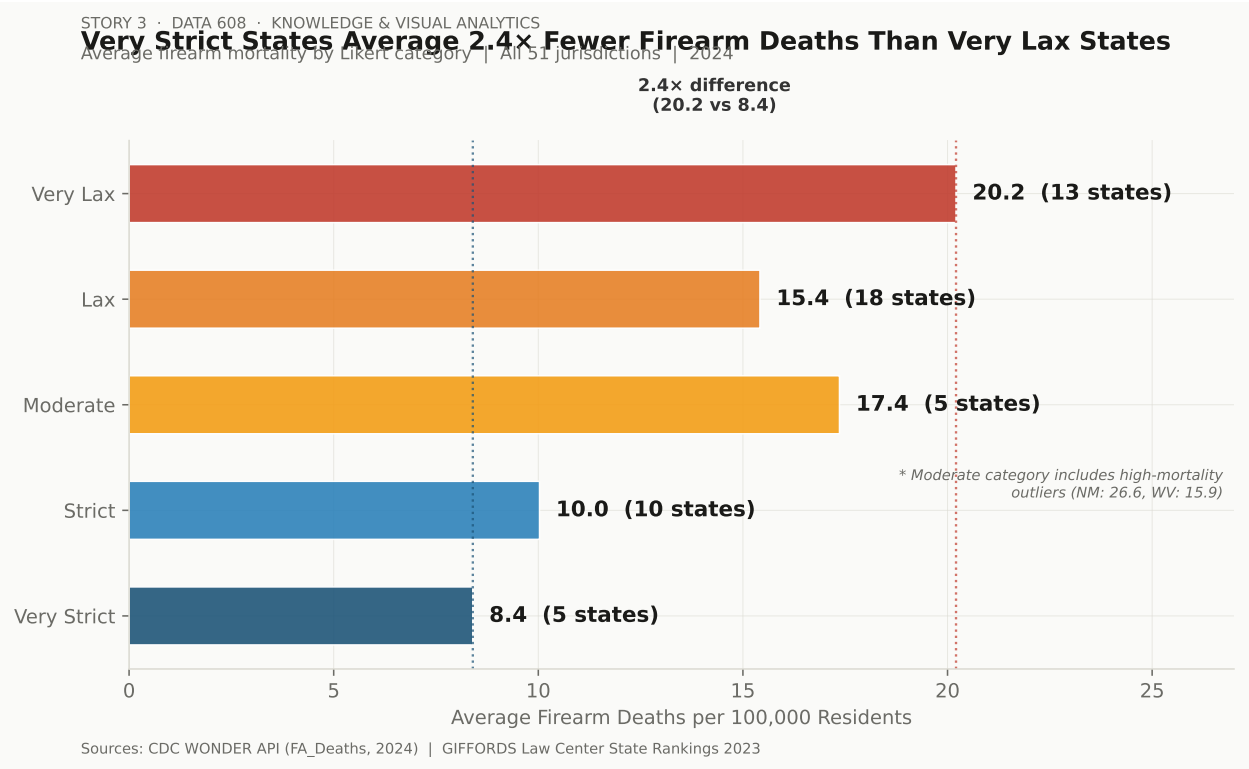

Figure 5: Average firearm mortality by gun law Likert category. The 2.4× gap between Very Lax (20.0) and Very Strict (8.4) is the clearest quantitative answer to the story's question.

Conclusion

The four visuals together tell a consistent story. Firearm mortality varies by 7× across U.S. states. That variation is not random — it follows a clear geographic and regulatory pattern. States with the strictest gun laws average **8.4 firearm deaths per 100,000 residents**; states with the most permissive laws average **20.0** — a **2.4× difference**. The scatter plot confirms this is not driven by a few outliers: the negative trend ($r = -0.67$) holds across all 51 jurisdictions.

The one irregularity — the Moderate category averaging slightly higher than Lax — is explained by two high-mortality states (New Mexico and West Virginia) that sit in a small category. It does not undermine the overall direction of the evidence.

The answer to the assignment question: Based on 2024 CDC data and GIFFORDS law rankings, stricter state gun laws are **strongly associated** with lower firearm mortality. The pattern is consistent, large in magnitude, and visible at every level of analysis — state-by-state, geographically, and in aggregate. Whether this association is causal requires controlling for confounding factors (poverty, urbanization, mental health access) — but the observational evidence is clear and substantial.

References

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